

268. Missing Number:

Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return *the only number in the range that is missing from the array*.

Example 1:

Input: `nums = [3,0,1]`

Output: 2

Explanation:

`n = 3` Since there are 3 numbers, so all numbers are in the range `[0, 3]`. 2 is the missing number in the range since it does not appear in `nums`.

Example 2:

Input: `nums = [0,1]`

Output: 2

Explanation:

`n = 2` Since there are 2 numbers, so all numbers are in the range `[0, 2]`. 2 is the missing number in the range since it does not appear in `nums`.

Example 3:

Input: `nums = [9,6,4,2,3,5,7,0,1]`

Output: 8

Explanation:

`n = 9` Since there are 9 numbers, all numbers are in the range `[0, 9]`. 8 is the missing number in the range since it does not appear in `nums`.

Constraints:

- `n == nums.length`
- `1 <= n <= 104`
- `0 <= nums[i] <= n`
- All the numbers of `nums` are **unique**.

Solution: Time = $O(n)$ and space = $O(1)$

```
class Solution:
    def missingNumber(self, nums: List[int]) -> int:
        n = len(nums)
        return n*(n+1)//2 - sum(nums)
```

Solution using XOR: Time = $O(n)$ and space = $O(1)$

```
class Solution:
    def missingNumber(self, nums: List[int]) -> int:
        res = 0
        for i, num in enumerate(nums):
            res ^= i ^ num
        return res ^ len(nums)
```